# LF1 Worked Examples: Average Rate of Change and Slope

Recording companion for the LF1 CheckIt support video — Math 172, Santa Ana College

## How to use this document

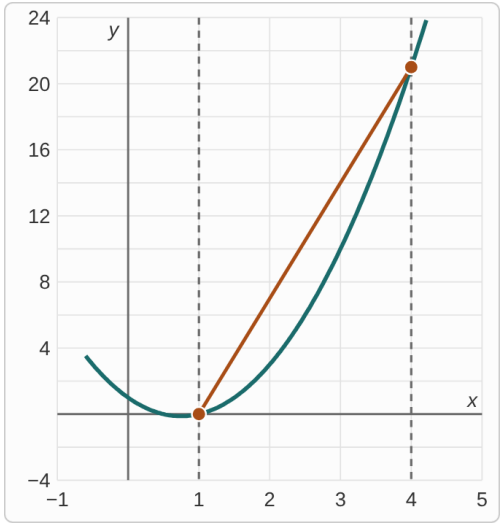
- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

## The one question behind every example

**How much did the output change, and how much did the input change?** Both average rate of change and slope are change in output divided by change in input. For a function on [a, b], that is f(b) − f(a) over b − a. For a line, use any two points: $y_2 - y_1$ over $x_2 - x_1$. Subtract in the same order on top and bottom, and divide by the real change in x, which is not always 1.

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# Example 1: Average Rate 1




**Plotted data:** A graph of the curve y = f(x) for x from about −0.6 to 4.2, with dashed vertical lines at x = 1 and x = 4. The curve dips just below the x-axis near x = 0.75 and then rises steeply. A straight segment joins the two points of the curve on the dashed lines, one on the x-axis at x = 1 and one high up at x = 4.

$$f(x)=2x^2-3x+1 \quad on \quad [1,4]$$

## Question

What is the average rate of change of f on [1, 4]?

- **A.** *21*
- **B.** *7*
- **C.** *1/7*
- **D.** *−7*

## Answer and reasoning

Find the outputs at the ends: $f(4)=2(16)-12+1=21$ and $f(1)=2-3+1=0$. Then divide the change in output by the change in input: $(f(4)-f(1))/(4-1)=(21-0)/3=7$. That is the slope of the straight segment in the picture. 21 is the change in output before dividing by 3; *1/7* divides the other way, change in input over change in output; and *−7* uses $f(1)-f(4)$ on top but $4-1$ on the bottom. **The average rate of change is 7.**

# Example 2: Average Rate 2

$$f(x)=x^4-2x \quad \text{on} \quad [-2,1]$$

## Question

What is the average rate of change of f on [−2, 1]?

- **A.** $-7$
- **B.** $-21$
- **C.** $11/3$
- **D.** $7$

## Answer and reasoning

$f(1)=1-2=-1$. At the negative end, $(-2)^4=16$, so $f(-2)=16-2(-2)=16+4=20$. The change in input is $1-(-2)=3$, so $(-1-20)/3=(-21)/3=-7$. $-21$ stops before dividing by the change in input; 7 subtracts the outputs in the other order on top only; and $11/3$ takes $(-2)^4$ as $-16$, so $f(-2)$ comes out $-12$. **The average rate of change is −7.**

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# Example 3: Average Rate 3

$$f(x)=\sqrt{(3x+2)} \quad on \quad [1,3]$$

## Question

What is the average rate of change of f on [1, 3]?

- **A.** $(\sqrt{11}-\sqrt{5})/2$
- **B.** $(\sqrt{6})/2$
- **C.** $3$
- **D.** $(\sqrt{11}+\sqrt{5})/2$

## Answer and reasoning

$f(3)=\sqrt{11}$ and $f(1)=\sqrt{5}$; neither simplifies. So the average rate of change is $(\sqrt{11}-\sqrt{5})/(3-1)=(\sqrt{11}-\sqrt{5})/2$, an exact answer to leave as it is. CheckIt may print it split up, as $1/2\sqrt{11}-1/2\sqrt{5}$, which is the same number. $(\sqrt{6})/2$ combines $\sqrt{11}-\sqrt{5}$ into $\sqrt{(11-5)}$, which is not allowed; 3 subtracts the numbers under the roots, $(11-5)/2$, and drops the roots; and + adds the outputs instead of subtracting them. ***$(\sqrt{11}-\sqrt{5})/2$*.**

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# Example 4: Average Rate 4

$$f(x)=(2x-1)/(x+2) \text{ on } [1,4]$$

## Question

What is the average rate of change of f on [1, 4]?

- **A.** *5/6*
- **B.** *18/5*
- **C.** *5/18*
- **D.** *−5/18*

## Answer and reasoning

*f(4)=7/6* and *f(1)=1/3=2/6*. The change in output is *7/6−2/6=5/6*, over a change in input of 3: *5/6÷ 3=5/18*. Fractions inside fractions are normal here; keep them exact. *5/6* stops before dividing by 3; *18/5* divides the other way; and *−5/18* subtracts the outputs in the other order on top only. ***5/18*.**

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# Example 5: Two Points 1

$$(-3,5) \text{ and } (1,-7)$$

## Question

What is the slope of the line through these points?

- **A.** *3*
- **B.** *−1/3*
- **C.** *−3*
- **D.** *1*

## Answer and reasoning

Change in *y*: *−7−5=−12*. Change in *x*, in the same order: *1−(−3)=4*. Slope: *(−12)/4=−3*. 3 subtracts the *y*'s in one order and the *x*'s in the other; *−1/3* puts the change in *x* on top; and 1 adds the coordinates, *(5+(−7))/(−3+1)*, instead of subtracting them. **The slope is *−3*.**

# Example 6: Two Points 2

$$(-4,6) \text{ and } (2,-8)$$

## Question

What is the slope of the line through these points?

**A.** $-7/3$
**B.** $-3/7$
**C.** $7/3$
**D.** $-7$

## Answer and reasoning

$(-8-6)/(2-(-4))=(-14)/6=-7/3$. Reduce the fraction, and leave it improper: CheckIt writes slopes like $-7/3$, not mixed numbers. $-3/7$ puts the change in $x$ on top; $7/3$ loses the sign, though $y$ goes down as $x$ goes up; and $-7$ divides by 2, the second $x$-value, instead of the change in $x$. **The slope is $-7/3$.**

# Example 7: Table 1

**Table 1.** Points on a line, listed in two rows

| **x** | −5 | −2 | 1 | 6 |
|---|---|---|---|---|
| **y** | 13 | 7 | 1 | −9 |

## Question

What is the slope of the line defined by this table?

**A.** $-6$
**B.** $-1/2$
**C.** $2$
**D.** $-2$

## Answer and reasoning

Use any two rows, such as $(-5,13)$ and $(-2,7)$: $(7-13)/(-2-(-5))=(-6)/3=-2$. Check with two others: $(-9-1)/(6-1)=(-10)/5=-2$. The $x$-values jump by 3, then 3, then 5, so the change in $x$ matters. $-6$ is the change in $y$ between the first two rows, as if $x$ went up by 1; $-1/2$ divides the other way; and 2 loses the sign, though $y$ goes down as $x$ goes up. **The slope is $-2$.**

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# Example 8: Table 2

**Table 2.** Points on a line, in columns

| x | y |
|---|---|
| −15 | 24 |
| −5 | 10 |
| 5 | −4 |
| 10 | −11 |

## Question

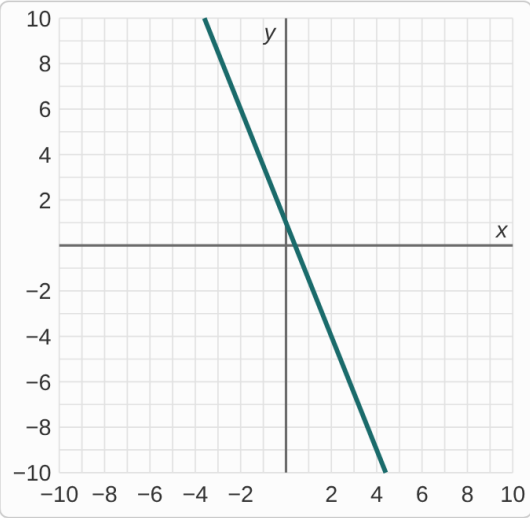
What is the slope of the line defined by this table?

**A.** *−14*
**B.** *−5/7*
**C.** *7/5*
**D.** *−7/5*

## Answer and reasoning

From *(−15,24)* to *(−5,10)*: *(10−24)/(−5−(−15))=(−14)/10=−7/5*. From *(5,−4)* to *(10,−11)*: *(−7)/5* again, even though that step in *x* is 5, not 10. *−14* is the change in *y* between the first two rows, without dividing by the change in *x*; *−5/7* divides the other way; and *7/5* loses the sign, though *y* goes down as *x* goes up. **The slope is *−7/5*.**

# Example 9: Graph 1




**Plotted data:** A straight line on a grid from −10 to 10 in both directions, with grid lines every 1 unit and no points marked. It falls steeply from left to right, passing through (−2, 6), (0, 1) and (4, −9).

## Question

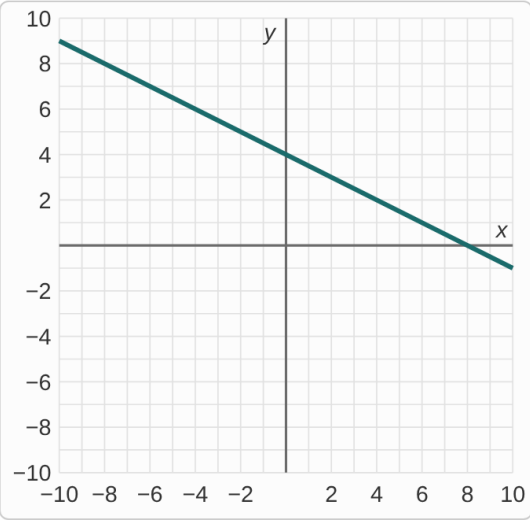
What is the slope of this line?

- **A.** *5/2*
- **B.** *−5/2*
- **C.** *−2/5*
- **D.** *1*

## Answer and reasoning

The line passes through *(0,1)*, but one step over it is not on a grid corner. The next corner is *(2,−4)*: over 2, down 5. So the slope is *(−5)/2=−5/2*. It falls from left to right, which rules out *5/2*. *−2/5* is run over rise; and 1 is where the line crosses the *y*-axis, not its slope. **The slope is *−5/2*.**

# Example 10: Graph 2




**Plotted data:** A straight line on a grid from −10 to 10 in both directions, with grid lines every 1 unit and no points marked. It falls gently from left to right, passing through (−10, 9), (0, 4) and (8, 0).

## Question

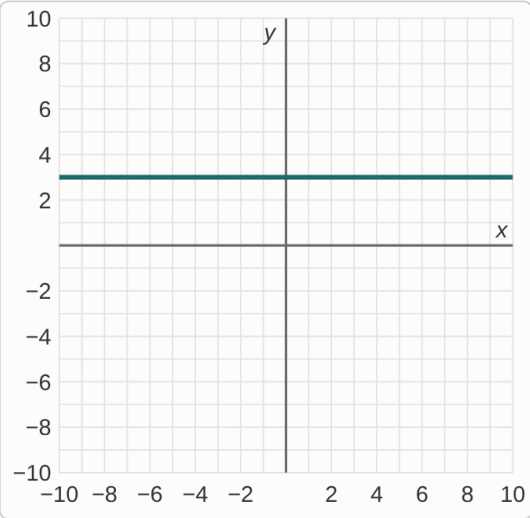
What is the slope of this line?

- **A.** *−2*
- **B.** *1/2*
- **C.** *4*
- **D.** *−1/2*

## Answer and reasoning

Grid corners at *(0,4)* and *(2,3)*: over 2, down 1, so *(−1)/2=−1/2*. It falls from left to right, so the slope is negative, which rules out *1/2*. *−2* is run over rise; and 4 is where the line crosses the *y*-axis. **The slope is *−1/2*.**

# Example 11: Graph 3




**Plotted data:** A straight line on a grid from −10 to 10 in both directions, with grid lines every 1 unit and no points marked. It runs straight across at height 3, passing through (−10, 3), (0, 3) and (10, 3).

## Question

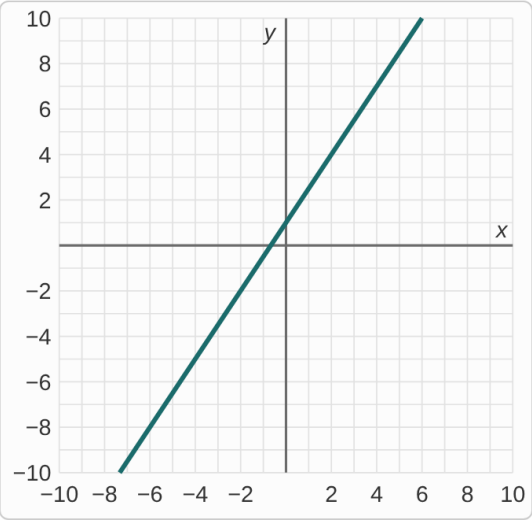
What is the slope of this line?

- **A.** *0*
- **B.** Undefined
- **C.** *3*
- **D.** *1*

## Answer and reasoning

Moving across, the height never changes: from *(0,3)* to *(5,3)* the rise is 0 and the run is 5, so *0/5=0*. A horizontal line has slope 0. A vertical line is the one with undefined slope, because its run is 0. 3 is the line's height, from its equation *y=3*, not its slope; and 1 counts one grid square per step without checking that the rise is 0. **The slope is 0.**

# Example 12: Graph 4




**Plotted data:** A straight line on a grid from −10 to 10 in both directions, with grid lines every 1 unit and no points marked. It rises from left to right, passing through (−6, −8), (0, 1) and (6, 10).

## Question

What is the slope of this line?

- **A.** *2/3*
- **B.** *3/2*
- **C.** *−3/2*
- **D.** *1*

## Answer and reasoning

The line passes through *(0,1)*, but the next grid corner is not one step over: it is *(2,4)*, up 3 and over 2. So the slope is *3/2*, a half-step slope like many on CheckIt. *2/3* is run over rise; *−3/2* has the wrong sign for a rising line; and 1 is where the line crosses the *y*-axis. **The slope is *3/2*.**

# Example 13: Slip 1

$$(2,5),\ (6,-3):\quad (-3-5)/(2-6)=(-8)/(-4)=2$$

## Question

A student found this slope. What is the mistake, if any?

**A.** There is no mistake.
**B.** The order switched: it should be −8/4 = −2.
**C.** Slope is run over rise, so it should be 1/2.
**D.** Add the coordinates instead, giving 2 over 8, or 1/4.

## Answer and reasoning

On top the student used the second point minus the first, $-3-5$, but on the bottom the first minus the second, $2-6$. Use one order for both: $(-3-5)/(6-2)=(-8)/4=-2$. The line falls from $(2,5)$ to $(6,-3)$, so its slope must be negative. Slope is rise over run, not run over rise, which would give $(-4)/(-8)=1/2$; and adding the coordinates, $(5+(-3))/(2+6)=2/8$, does not measure change at all. **The slope is $-2$.**

# Example 14: Slip 2

$$f(x)=x^2 \text{ on } [1,3]: \quad (f(3)-f(1))/(3-1)=(9-1)/2=4$$

## Question

A student found this average rate of change. What is the mistake, if any?

**A.** There is no mistake.
**B.** Stop at the change in output: f(3) − f(1) = 8.
**C.** Divide by 3, the right end of the interval: 8/3.
**D.** Average the two outputs instead: (9 + 1)/2 = 5.

## Answer and reasoning

*f(3)=9* and *f(1)=1*, so the change in output is 8 over a change in input of 2: *8/2=4*. The student is right. Stopping at 8 skips the division by the change in input; dividing by 3 uses the endpoint instead of the change *3−1*; and the average rate of change is not the average of the outputs. **There is no mistake.**

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# Example 15: Slip 3

$$y=-4$$

## Question

A student says the slope of the line y = −4 is undefined, because it has no x in it. What is the mistake, if any?

**A.** There is no mistake.
**B.** It should be −4, the number in its equation.
**C.** It is horizontal, so its slope is 0.
**D.** It should be 4, the distance from the line to the x-axis.

## Answer and reasoning

Every point on $y=-4$ has height $-4$, so the line is horizontal: the rise between any two points is 0, and $0/(\text{run})=0$. Undefined slope belongs to vertical lines, like $x=2$, whose run is 0. $-4$ is the line's height, and 4 is its distance from the $x$-axis; neither is a slope. **The slope is 0.**

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# Example 16: Slip 4

**Table 3.** A student's table of points on a line

| x | 0 | 3 | 6 | 9 |
|---|---|---|---|---|
| y | 2 | 8 | 14 | 20 |

## Question

A student says the slope is 6, because y goes up by 6 each time. What is the mistake, if any?

**A.** There is no mistake.
**B.** Slope is change in x over change in y: 3/6 = 1/2.
**C.** Divide the last row instead, so it is 20/9.
**D.** x goes up by 3 each time, so it is 6/3 = 2.

## Answer and reasoning

*y* does go up by 6 each time, but *x* goes up by 3, not 1. So the slope is *6/3=2*: 2 up for each 1 over. Change in *x* over change in *y* is upside down; and *20/9* divides one point's coordinates, which only works for a line through the origin, and this one starts at height 2. **The slope is 2.**

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# Example 17: Slip 5

$$f(x)=\sqrt{x} \text{ on } [4,9]: \quad (\sqrt{9}-\sqrt{4})/(9-4)=(\sqrt{5})/5$$

## Question

A student found this average rate of change. What is the mistake, if any?

**A.** There is no mistake.
**B.** $\sqrt{9} - \sqrt{4}$ is $3 - 2 = 1$, so it is $1/5$.
**C.** Divide by $\sqrt{5}$ instead, which gives 1.
**D.** The interval length is 9, so divide $\sqrt{5}$ by 9.

## Answer and reasoning

The student turned $\sqrt{9}-\sqrt{4}$ into $\sqrt{(9-4)}=\sqrt{5}$, but roots do not subtract that way. Evaluate each one first: $\sqrt{9}-\sqrt{4}=3-2=1$, so the answer is $1/5$. The change in input is $9-4=5$, not $\sqrt{5}$ or 9. **The average rate of change is $1/5$.**

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